

KAUSTUBH WATANE

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SUMMARY

AI Engineer at GE HealthCare who shipped three production AI systems in 10 months — solo, on a regulated medical-device network. The agentic AI workspace was adopted by over 50% of the engineering team and drove 30%+ efficiency gains. Full-stack ownership across architecture, backend, frontend, and deployment. Builds AI that ships and stays shipped.

WORK EXPERIENCE

AI Engineer (Trainee), GE HealthCare, Bangalore

Aug 2025 - Present

Nebula AI — Enterprise AI Workspace

- Designed and built Nebula AI, a secure, on-premise retrieval-augmented generation (RAG) assistant for GE HealthCare using FastAPI, PostgreSQL, Qdrant, and Ollama, combining hybrid search (BM25 and dense embeddings) with reranking to deliver accurate, source-grounded answers from internal documents.
- Built a real-time voice AI agent with speech-to-text and text-to-speech that lets users create tasks, notes, and calendar events hands-free, with sub-1-second response latency.
- Expanded the platform into a full AI workspace, adding a Code Studio for AI-generated code, smart notes with auto-filled templates, and an admin panel for access control and audit logging — driving 50%+ team adoption and a 30%+ improvement in efficiency as the sole engineer owning design, development, and deployment.

TestMatrix AI — V&V Test Automation

- Auto-generates IEC 62304 / FDA 21 CFR 820 compliant test cases for medical-device hardware — output enters human review ready to approve, not ready to fix.
- Layered quality pipeline ensures cheap checks always run before expensive AI calls, keeping output consistent.

Atlas AI — Engineering Document Generator

- Generates 50+ page DRS/SSRS/SWRS specification documents from a short brief in under 10 minutes — work that previously took engineers weeks to complete manually.
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TECHNICAL SKILLS

- **Programming:** Python, TypeScript, JavaScript, SQL, Dart, Embedded C
- **Generative AI / LLM:** Retrieval-Augmented Generation (RAG), AI Agents / Agentic AI, Multi-Agent Systems, LangChain, LangGraph, Prompt Engineering, Tool/Function Calling, LLM Evaluation, Guardrails, Hallucination Detection
- **Retrieval & NLP:** Embeddings, Semantic Search, Hybrid Search (BM25 + Dense), Reranking, Vector Databases (Qdrant), Document Parsing / OCR
- **Voice AI:** Speech-to-Text (STT), Text-to-Speech (TTS), Voice Activity Detection, Real-Time Streaming, WebSockets
- **Backend & Data:** FastAPI, REST APIs, SQLAlchemy, Pydantic, Node.js, PostgreSQL, Firebase
- **Frontend / Mobile:** React, React Native, Expo, Flutter, Next.js
- **Cloud & LLMOps:** Docker, Azure AI Services, Cloud AI/ML Deployment, MLOps, Ollama, On-Premise Inference, Responsible AI

PROJECTS

Luna AI — AI Companion Platform

- Built and deployed a cross-platform AI companion app using React Native and Expo, with a FastAPI + PostgreSQL backend, integrating Gemini, Claude, Grok, and OpenAI APIs through a unified conversational interface.
- Implemented AI chat, voice interactions, journaling, games, and personalized companion features; conversation state and user data persisted via PostgreSQL.
- Independently architected the full stack — React Native frontend, FastAPI backend, API integration layer, and Play Store deployment pipeline — achieving 500+ users post-launch.

Spenzy AI — Smart Expense Tracker

- Developed a cross-platform expense tracking app using Flutter (Dart) with OpenAI API integration for AI-powered financial assistance and natural-language expense analysis.
- Implemented expense logging, spending categorization, budget tracking, and conversational financial summaries through REST API integration with GPT.
- Designed responsive mobile UI in Flutter with real-time spending insights and personalized budget recommendations.

CERTIFICATIONS

- **Docker Foundations Professional Certificate** — containerization, Docker Compose, image optimization
- **Microsoft Azure AI Essentials Professional Certificate** — Azure AI services, cloud AI/ML deployment, responsible AI
- **Building AI Products: Architecture and Orchestration Professional Certificate** — AI system architecture, agent orchestration, LLM application design

EDUCATION

PG Diploma In Embedded System and Design	Aug 2024 - March 2025
CDAC ACTS, Pune	Grade A
Bachelor of Electronics and Telecommunication Engineering	June 2019 - June 2023
Government College of Engineering, Karad	CGPA 7.83/10